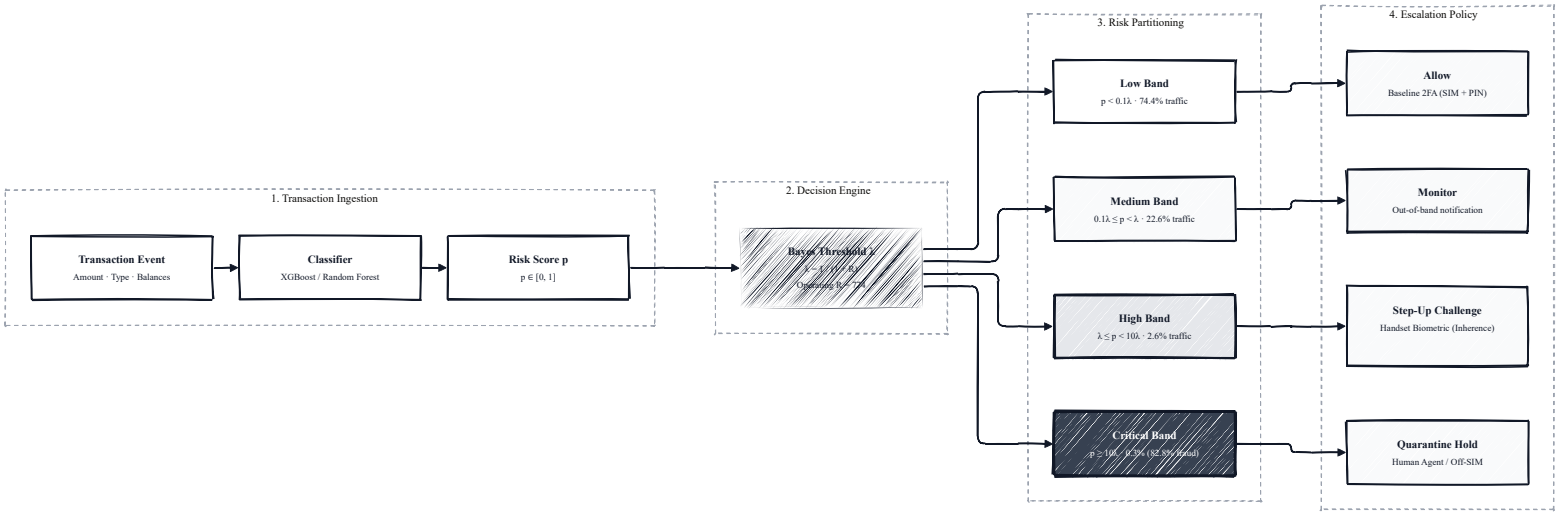

1. Transaction Ingestion
Transaction Event
Amount · Type · Balances
Classifier
XGBoost / Random Forest
Risk Score p
p ∈ [0, 1]
2. Decision Engine
Bayes Threshold λ*
λ* = 1 / (1 + R)
Operating: R = 74
3. Risk Partitioning
Low Band
p < 0.1λ · 74.4% traffic
Medium Band
0.1λ ≤ p < λ · 22.6% traffic
High Band
λ ≤ p < 10λ · 2.6% traffic
Critical Band
p ≥ 10λ · 0.3% (82.8% fraud)
4. Escalation Policy
Allow
Baseline 2FA (SIM + PIN)
Monitor
Out-of-band notification
Step-Up Challenge
Handset Biometric (Inherence)
Quarantine Hold
Human Agent / Off-SIM